

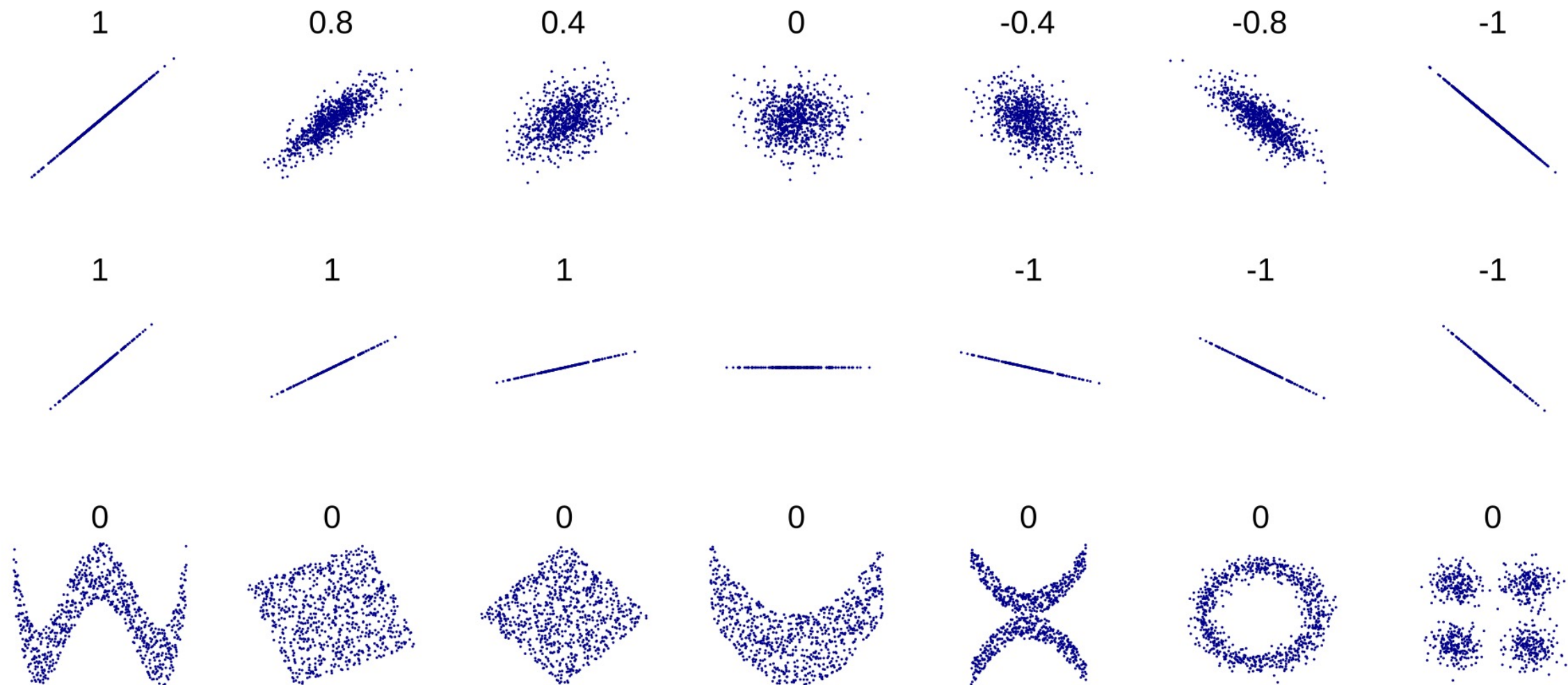
# Lecture 4 — Correlation & Regression

Michal Kolář

IMG

Prague, 06/2026

# Part I — Measuring association



# Correlation: Pearson's r

How strongly do two measured variables move together?

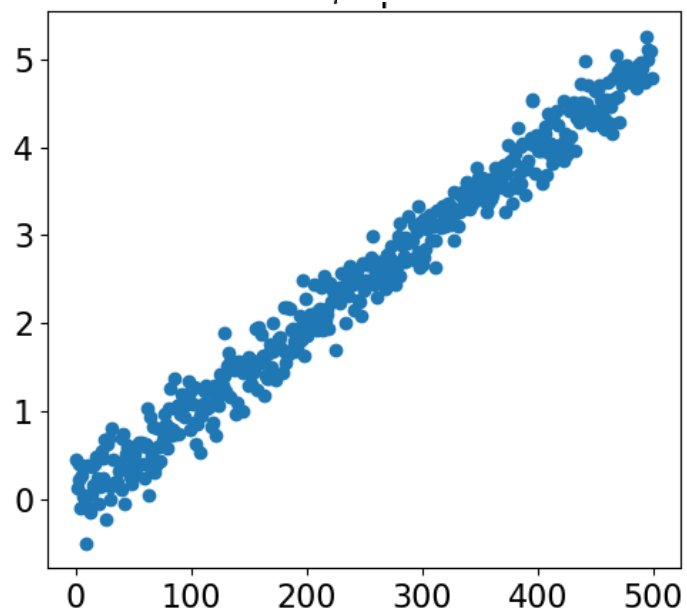
- Correlation runs from  $-1$  [anti] through  $0$  [unrelated] to  $+1$ ; a strong correlate is a **predictor**, carrying redundant information.
- **Pearson's r** is a standardized covariance — it measures **linear** association only

$$r_{x,y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$

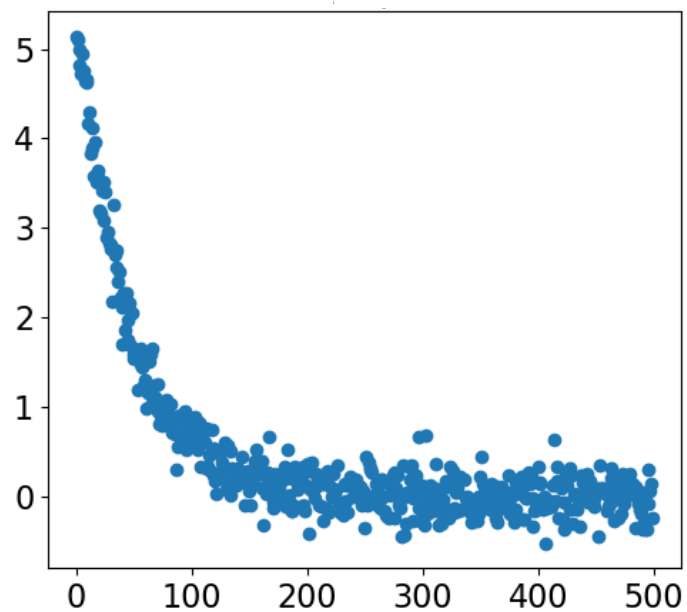
- Pitfall: a strong non-linear relationship can still give  $r \approx 0$  — **always plot the data**.

*$r = 0$  means no linear association — not independence.*

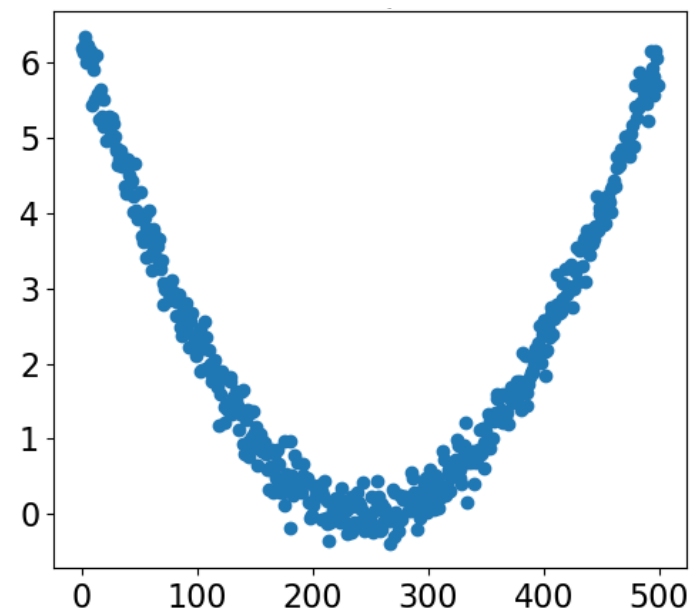
Pearson: 0.989,



Pearson: -0.660



Pearson: -0.007



# When linear isn't enough: Spearman & Kendall

Rank-based correlations for monotonic relationships.

- Spearman's  $\rho$  = Pearson computed on the ranks  $\rightarrow$  captures any monotonic relationship.

$$\rho = 1 - \frac{6 \sum d_i^2}{n^3 - n}$$

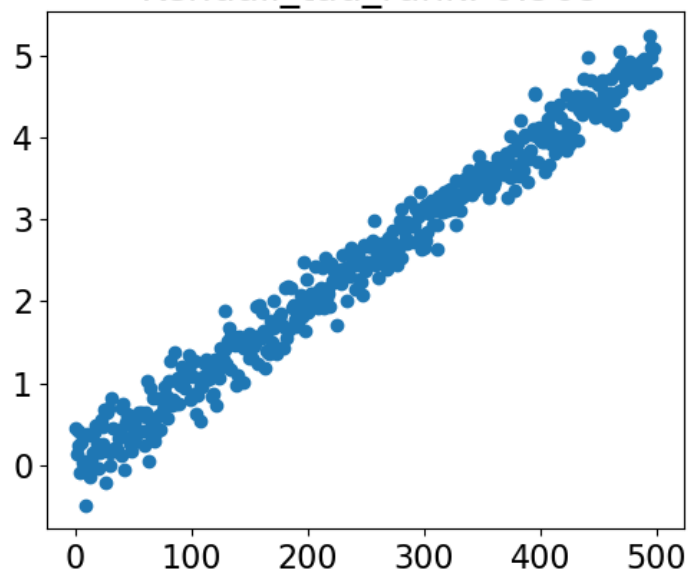
- $d_i \equiv R[X_i] - R[Y_i]$  is the difference between the two ranks of each observation
- Kendall's  $\tau$  = concordant [C] minus discordant [D] pairs, normalized.

$$\tau = \frac{C - D}{n(n-1)/2}$$

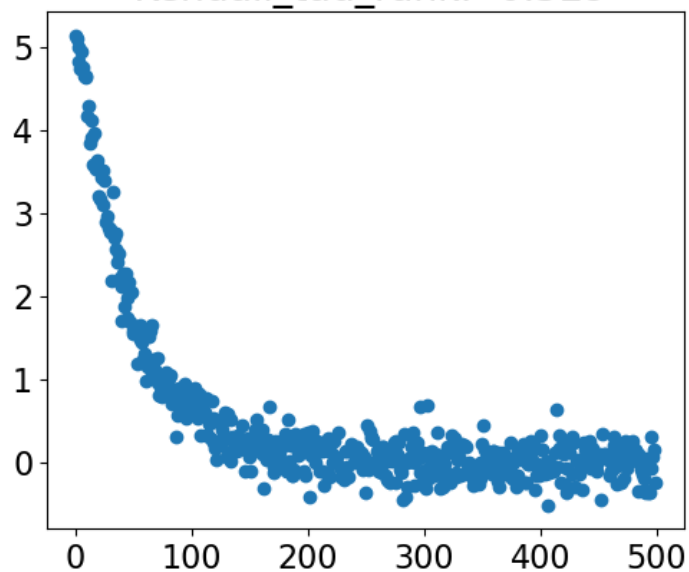
- Because they use only order, Spearman and Kendall are far less sensitive to outliers and scale.

*Ordinal, monotonic-but-non-linear, or outlier-prone  $\rightarrow$  rank methods.*

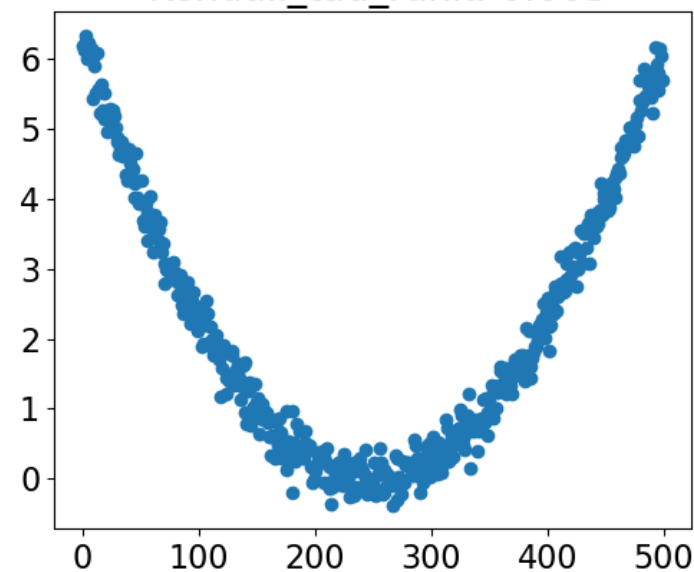
Pearson: 0.989, Spearman: 0.990  
Kendall\_tau: 0.909,  
Kendall\_tau\_rank: 0.909



Pearson: -0.660, Spearman: -0.692  
Kendall\_tau: -0.529,  
Kendall\_tau\_rank: -0.529



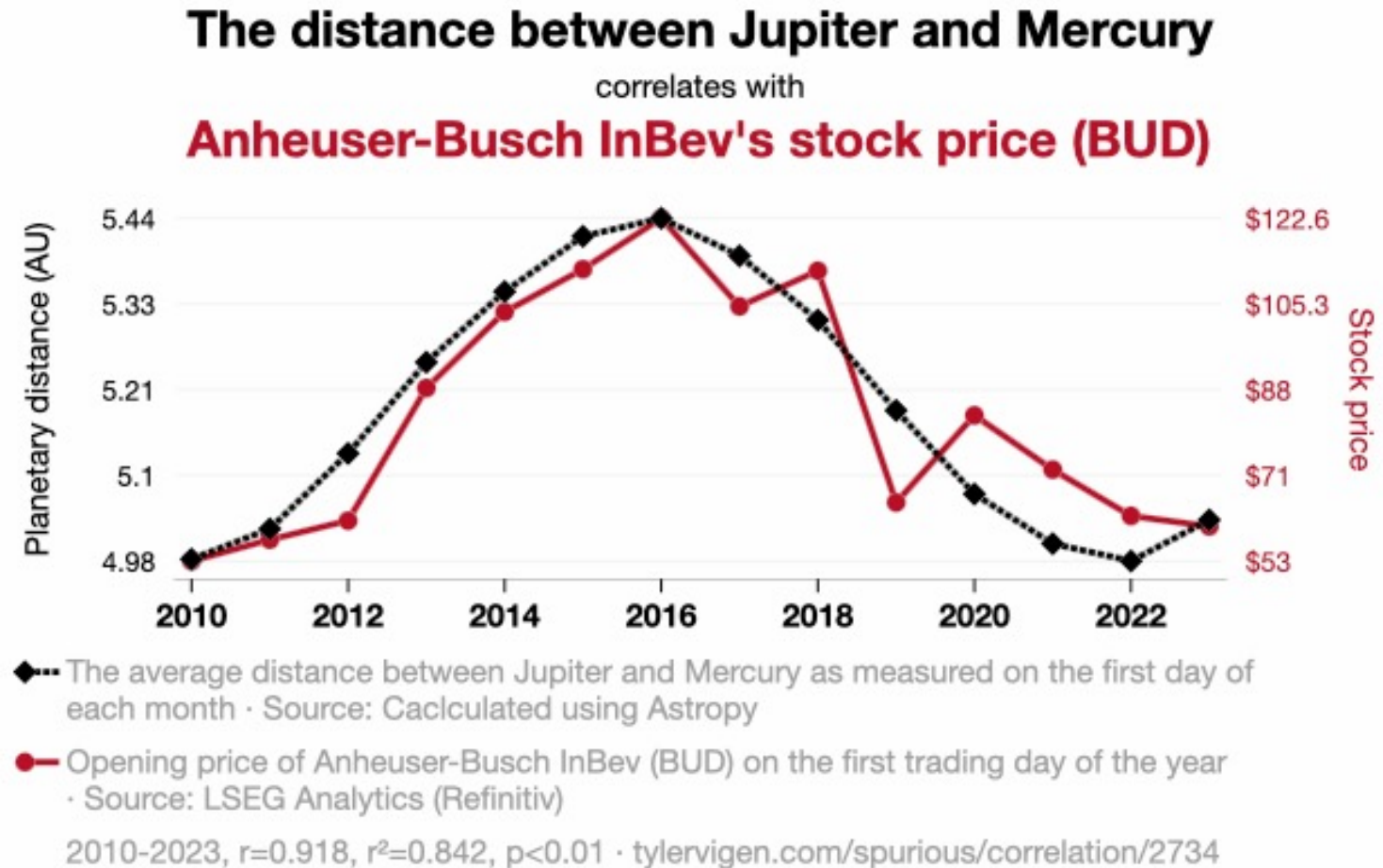
Pearson: -0.007, Spearman: -0.002  
Kendall\_tau: 0.003,  
Kendall\_tau\_rank: 0.003



# Significance, and the two big caveats

Is the correlation real — and what does it actually mean?

- Test  $H_0: \rho = 0$  via a **t-test**, a **permutation test**, or **Fisher's arctanh transform**.
- **Correlation  $\neq$  causation**: nothing about direction, nor whether a hidden confounder drives both.
- Spurious correlations are everywhere — apply a critical eye and visualize.



*A fitted number is not automatically a causal effect.*



## Part II — Modelling a relationship

# REGRESSION

- Technique used for the modeling and analysis of numerical data
- Exploits the relationship between two or more variables so that we can gain information about one of them through knowing values of the other
- Regression can be used for prediction, estimation, hypothesis testing, and modeling causal relationships

$$Y = f[X_1, X_2]$$

- $Y$  is a dependent variable [or outcome variable, or response variable]
- $X$ 's are independent variables [or predictor variables, or explanatory variables]

# LINEAR REGRESSION

- Suppose we want to model the dependent variable  $Y$  in terms of two predictors,  $X_1, X_2$

$$Y = f[X_1, X_2]$$

- Typically, we will not have enough data to try to directly estimate  $f$
- Therefore, we usually have to assume that  $f$  has some restricted form, such as linear

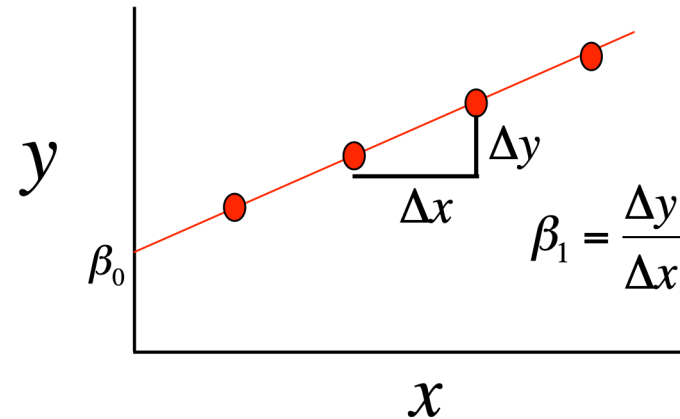
$$Y = a + bX_1 + cX_2$$

*Correlation is one number; regression is a full predictive model.*

# LINEAR REGRESSION IS A PROBABILISTIC MODEL

- Much of mathematics is devoted to studying variables that are deterministically related to one another

$$y = \beta_0 + \beta_1 x$$



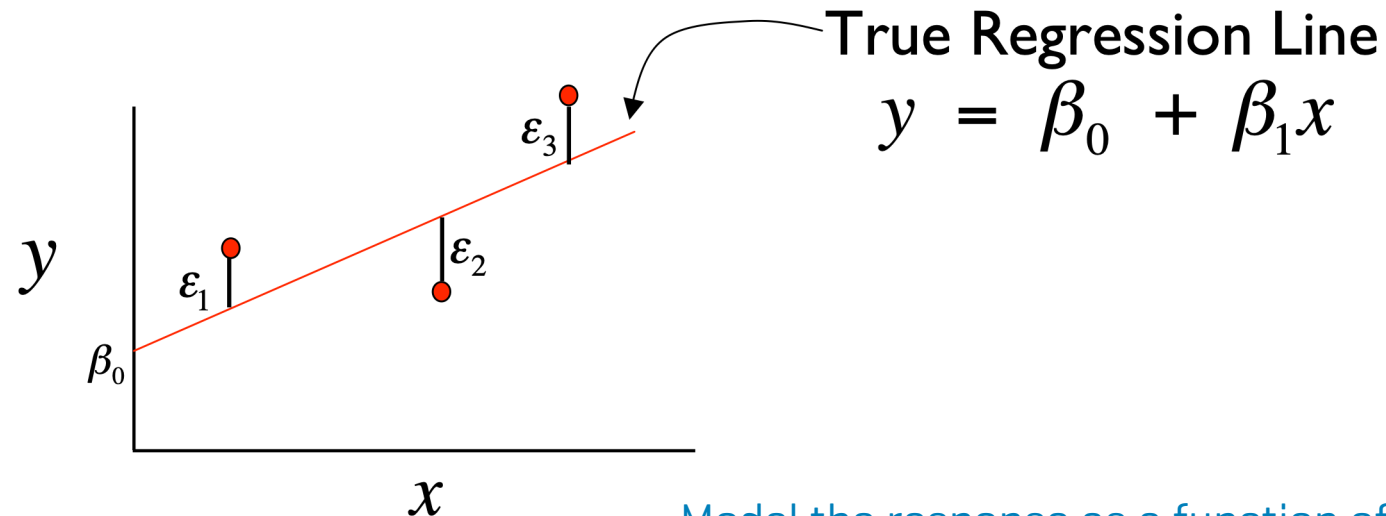
- But we're interested in understanding the relationship between variables related in a nondeterministic fashion

# LINEAR REGRESSION IS A PROBABILISTIC MODEL

- Definition: There exists parameters  $\beta_0$ ,  $\beta_1$ , and  $\sigma^2$  such that for any fixed value of the independent variable  $x$ , the dependent variable  $y$  is related to  $x$  through the model equation

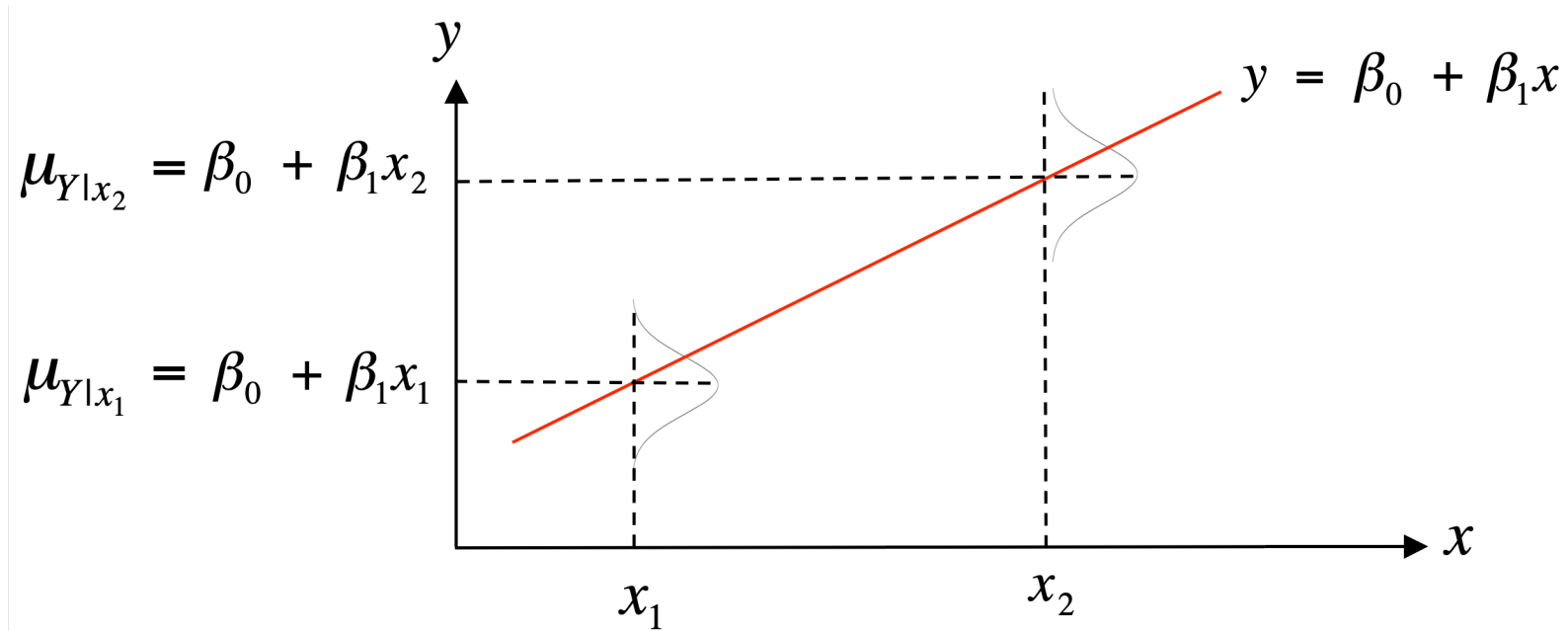
$$y = \beta_0 + \beta_1 x + \epsilon$$

- $\epsilon$  is a random variable assumed to be  $\mathcal{N}(0, \sigma^2)$



Model the response as a function of predictors plus noise.

# GRAPHICAL INTERPRETATION



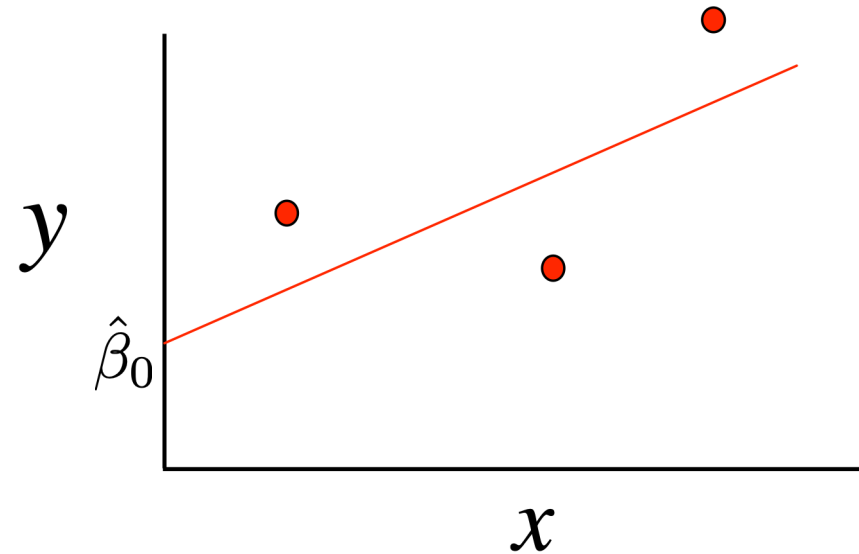
For example if  $y$  is log expression intensity of a gene and  $x$  is age then  $\mu_{Y|2}$  is the average expression intensity in 2 year old children.

We need homoscedasticity

# ORDINARY LEAST SQUARES [OLS] PARAMETER ESTIMATION

- Point estimates of  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are obtained by the principle of least squares

$$S(\beta_0, \beta_1) = \sum_i [y_i - (\beta_0 + \beta_1 x_i)]^2$$



*Squared residuals → sensitive to outliers*

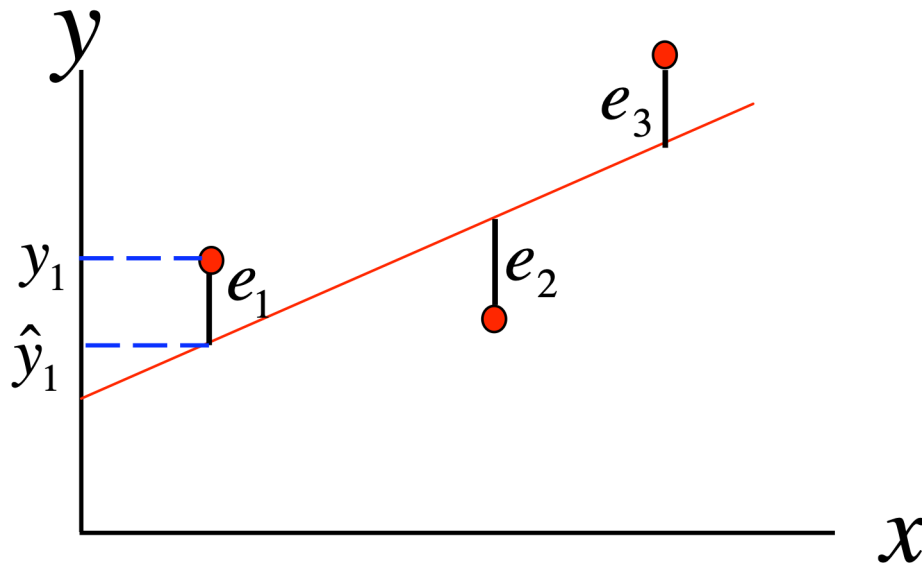
# PREDICTED AND RESIDUAL VALUES

- Predicted, or fitted, values are values of  $y$  predicted by the least-squares regression line obtained by plugging in  $x_1, x_2, \dots, x_n$  into the estimated regression line

$$\hat{y}_j = \hat{\beta}_0 + \hat{\beta}_1 x_j$$

- Residuals are the deviations of observed and predicted values

$$e_j = y_j - \hat{y}_j$$





# RESIDUALS ARE USEFUL

- They allow us to calculate the error sum of squares [SSE]:

$$SSE = \sum_j (e_j)^2 = \sum_j (y_j - \hat{y}_j)^2$$

- Which in turn allows us to estimate  $\sigma^2$  [ $n$  is the number of observations]:

$$\hat{\sigma}^2 = \frac{SSE}{n - 2}$$

- As well as an important statistic referred to as the coefficient of determination

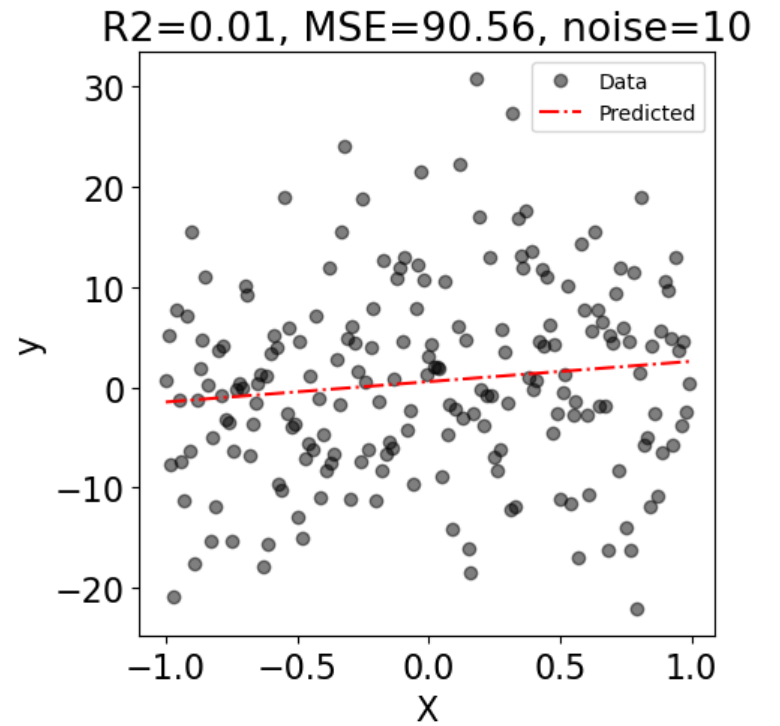
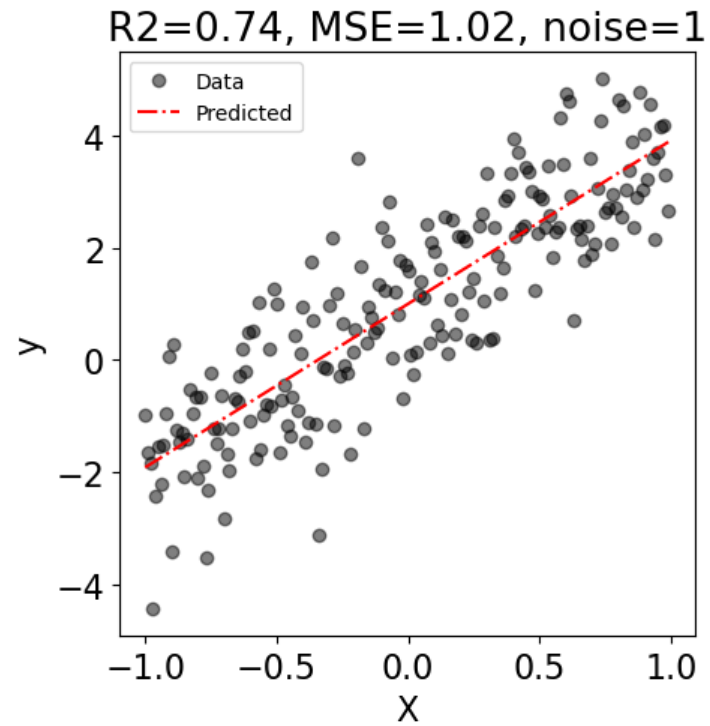
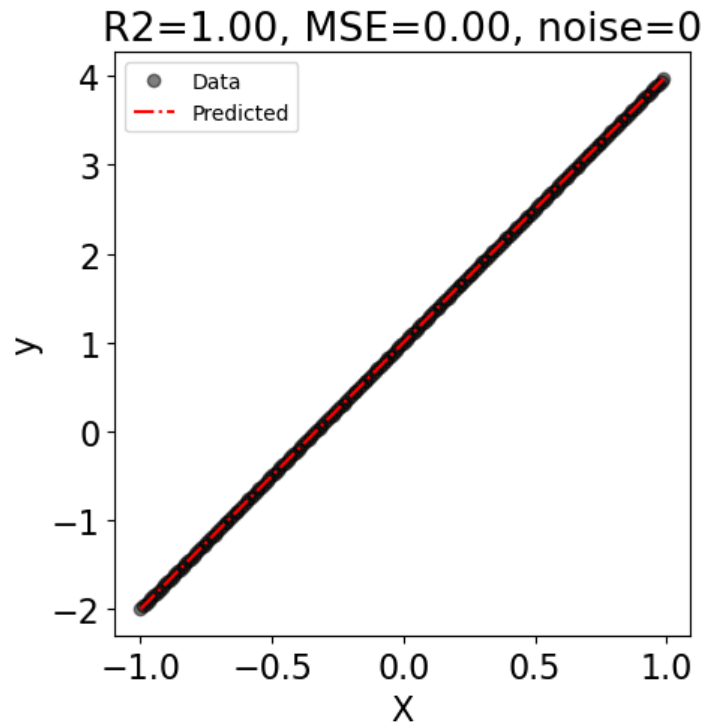
$$R^2 = 1 - \frac{SSE}{SST}; \quad SST = \sum_j (y_j - \bar{y})^2$$

- Or its adjusted variant

$$R_a^2 = 1 - \frac{n-1}{n-p} (1 - R^2)$$

*Plain  $R^2$  never decreases as predictors are added — hence adjusted  $R^2$ .*

# RESIDUALS ARE USEFUL



# IS THE EXPLANATORY VARIABLE EXPLANATORY?

- we have fitted a model:  $y = \beta_0 + \beta_1 x + \epsilon$
- significance of influence of X on Y translates to significance of  $\hat{\beta}_1$
- Hypothesis tests for  $\hat{\beta}_1$  can be done by simple t-tests:

$$H_0 : \hat{\beta}_1 = 0$$

$$H_A : \hat{\beta}_1 \neq 0$$

$$t = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)}$$

- where  $t$  follows an approximate t-distribution with  $[n - 2]$  degrees of freedom [ $n$  observations, two parameters]
- in general  $se(X) = c \sigma(X)$  and thus

$$t = \frac{\hat{\beta}_1}{s_1 c}$$

# INTERMEZZO

- for one sample t-test

$$c = \sqrt{\frac{1}{n}}$$

- for two sample t-test

$$c = \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

- for linear regression

$$c = \sqrt{\frac{1}{\sum_j (x_j - \bar{x})^2}}$$

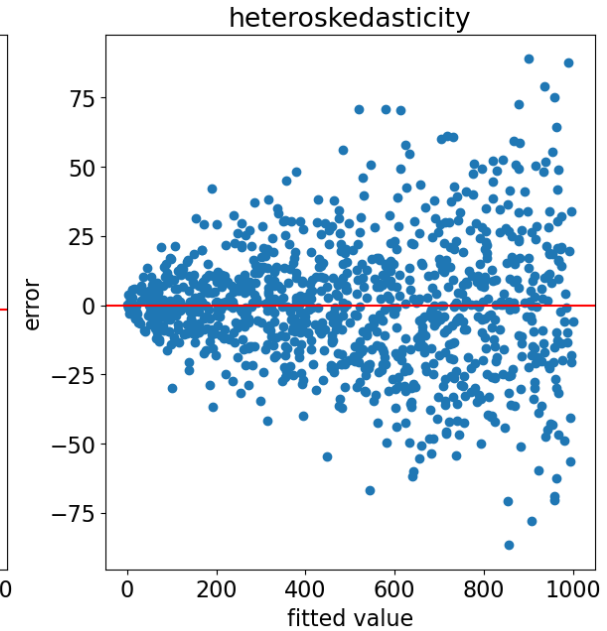
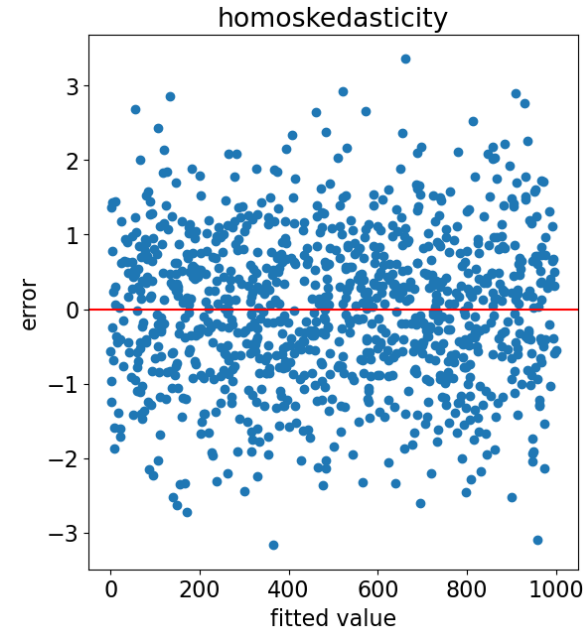
Always, it is some value independent from the response variable.

It shows depth or breadth of the data

# OLS assumptions

The fit is easy; trusting it needs assumptions.

- Correct **specification** and strict **exogeneity** (errors centred on the truth).
- No **multicollinearity**: redundant predictors make coefficients non-identifiable.
- **homoscedasticity** (constant spread) and **no autocorrelation**.
- Normality is **not needed for the fit** — but it is needed for CIs and tests.



*Normal errors  $\Rightarrow$  OLS coincides with Maximum Likelihood.*

## Part III — Comparing & extending models

# Maximum likelihood

The general estimation principle behind OLS.

- **Likelihood** = how probable the data are under some model parameters  $\Theta$ ; maximise the log-likelihood.

$$l(\Theta; Y) = \sum_i \ln p(y_i | \Theta)$$

- Under Gaussian noise, maximising likelihood = minimising squared residuals  $\rightarrow$  **ML  $\equiv$  OLS**.
- Parametric: requires an assumed density — and the gateway to GLMs.

*Assume normal errors and ML hands you least squares.*

# Comparing models

Is the extra parameter worth keeping?

- Two models are **nested** if you can get the simpler one by switching off (fixing to zero) some parameters of the bigger one
- **Likelihood Ratio Test** — for nested models;  $\sim \chi^2$  with df = the parameter difference.

$$\text{LRT} = 2(l_1 - l_0) \sim \chi^2_{\Delta df}$$

- **Akaike information criterion [AIC] or Bayesian information criterion [BIC]** — compare non-nested models; lower is better, but no clean p-value.

$$\text{AIC} = 2k - 2\hat{l} \quad \text{BIC} = k \ln n - 2\hat{l}$$

[k = parameters,  $n$  = observations,  $l$  = maximised log-likelihood]

- Both guard against **overfitting**: more parameters always fit the training data better.

*LRT: does adding this parameter beat chance?*



# When OLS fails, and regularization

Fixes for broken assumptions — and shrinking complexity.

- **Heteroscedasticity** → transform the data, or Weighted / Generalised Least Squares.
- **Regularization** penalises coefficient size: **Lasso** [L1] can zero predictors; **Ridge** [L2] shrinks them.

$$S(\beta_0, \beta_1) = \sum_i [y_i - (\beta_0 + \beta_1 x_i)]^2$$

$$\text{Ridge: } S(\beta) + \frac{1}{C} \sum_i \beta_i^2$$

$$\text{Lasso: } S(\beta) + \frac{1}{C} \sum_i |\beta_i|$$

- Choosing the penalty strength  $C$  is **machine learning** [cross-validation] — beyond this course.

*Lasso selects features; Ridge shrinks smoothly.*